VIJAYA SIVANJAN KOMMURI

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PROFILE

Research-oriented AI & Machine Learning engineer with an IEEE conference submission (SPICES 2026), a published journal paper, and 33+ credits in AI/ML coursework. Built a calibrated ensemble system achieving AUC-ROC 0.967 for power outage prediction with open-source full-stack deployment. Six months of production full-stack development at VTAG Software. Seeking M.Sc. in AI, Computer Science, or Autonomous Systems in Germany.

RESEARCH & PUBLICATIONS

[1] V.S. Kommuri, T. Mahadev, R.C.P. Chase, Dr. Md. Adam Baba. *"Compound Weather Event Interaction Modeling with Calibrated Uncertainty for State-Agnostic Power Outage Prediction."* Submitted to 5th IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES) 2026. Paper ID: 822. Status: Under Review.

Key results: AUC-ROC 0.967 | F1 0.947 | ECE 0.004 | Cross-state generalization gap <0.007

[2] *"Adaptive Graph Neural Networks with Curriculum Learning for Robust Social Network Analysis."* Published in IJSREM, Vol. 09, Issue 08, August 2025. ISSN: 2582-3930.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

Malla Reddy University, Hyderabad | Oct 2021 – Jun 2026 (Expected)

CGPA: 8.09/10 (German equivalent: ~2.0 "Gut") | APS-verified | Credits: 166+ | EQF Level 6

Academic Credit Distribution

Domain	Credits	Key Subjects	Avg Grade
AI & Machine Learning	33	Deep Learning, NLP, Generative AI, Information Retrieval, ML Lab	A (9.0)
Core Computer Science	42	Algorithms, Automata/Compilers, OS, DBMS, Full-Stack Web, Cloud	B+ (7.8)
Mathematics & Statistics	13	Calculus, Linear Algebra, Probability, Discrete Math	B+ (7.0)
Edge Computing & IoT	9	Edge Cloud Computing, IoT Applications	A (8.7)
Software Engineering	9	Software Quality & Testing, OO Software Eng	A (8.3)
Bachelor Thesis	12	Compound Weather Event Modeling (IEEE SPICES 2026)	In Progress

Grade Trajectory: *Sem 3: 7.23 → Sem 5: 8.94 → Sem 7: 8.67 (sustained post-recovery improvement after health-related intermission)*

Class XII: 90.2% (Narayana Junior College) | Class X: 9.5 CGPA (Ravindra Bharathi School)

BACHELOR THESIS PROJECT

Compound Weather Event Modeling for State-Agnostic Power Outage Prediction

Supervisor: Dr. Md. Adam Baba | Duration: Jan 2026 – Jun 2026

- Developed ML framework predicting weather-induced power outages 24h ahead, introducing novel compound weather event features (co-occurrence matrices, pairwise severity products, sequential escalation) across 6 weather categories
- Engineered 138 features from 1,945 geocoded NOAA Storm Events on Uber H3 hexagonal grid (~5.16 km²) covering 12,000 samples across Texas, California, and Florida
- Built calibrated XGBoost/LightGBM ensemble: AUC-ROC 0.967 (95% CI: 0.944–0.984), F1 0.947, perfect precision at 0.90 recall, ECE reduced 98.4% via isotonic calibration
- Designed state-agnostic YAML config architecture — cross-state AUC-ROC gap only 0.002–0.007
- Full-stack deployment: FastAPI + Next.js 14 + TimescaleDB + Redis + MLflow + Docker Compose | 57 automated tests
- Open-source: github.com/DarthAether/outage-prediction-system

PROFESSIONAL EXPERIENCE

Full-Stack Software Developer Intern

VTAG Software, Hyderabad | Dec 2025 – Jun 2026

- Built broker dashboard, property management workflows, and admin approval systems for production real-estate platform using TypeScript, Next.js, React, NestJS, GraphQL, Prisma, PostgreSQL
- Developed backend for Senior Living platform: RESTful APIs, JWT auth, modular service architecture, Express.js, Prisma
- Authored E2E automated test suites using Playwright across interconnected frontend-backend workflows
- Built lead attribution, campaign-linked sharing features with tokenized links and AWS S3 media automation

ADDITIONAL PROJECT

IoT Guard — Smart Home Security Tool

- Desktop security application using PySide6 + Google Gemini API for intelligent threat analysis and natural language security log summaries

TECHNICAL SKILLS

Domain	Technologies
AI & Machine Learning	Python, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SHAP, GANs, VAEs, Transformers, NLP, Computer Vision, Uncertainty Quantification
Full-Stack Development	TypeScript, JavaScript, Next.js 14, React, NestJS, Express.js, GraphQL, Apollo, Prisma, FastAPI, Tailwind CSS
Databases & Infrastructure	PostgreSQL, TimescaleDB, Redis, AWS S3, MLflow, Docker, Docker Compose
Tools & Methods	Git, Playwright (E2E Testing), H3 Spatial Indexing, Isotonic Calibration, SHAP Explainability, Linux

LANGUAGE SKILLS

Language	Level
English	C1–C2 (IELTS 8.0: L 8.5, R 8.5, W 7.5, S 7.5)
Hindi	B2–C1
Telugu	Native
German	A1 (actively learning)

CERTIFICATIONS

- IELTS Academic: Band 8.0 — August 2025
- APS Certificate: Obtained December 2025, Direkter Zugang